MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-16B/U

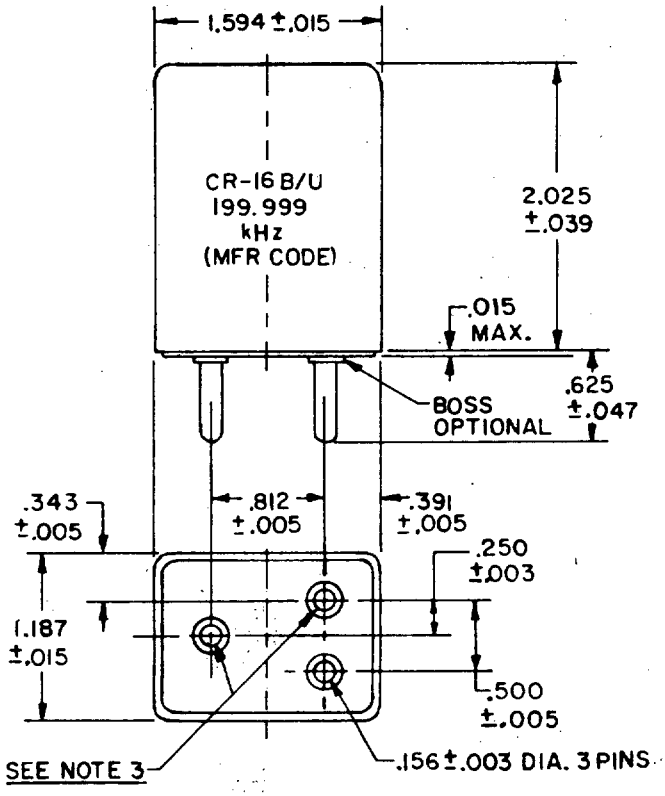
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of specification MIL-C-3098.

Pertinent characteristics: 75 to 200 kHz; fundamental; noncontrolled; series.

Ⓔ THIS SPECIFICATION INACTIVE FOR NEW DESIGN

INCHES	MM	INCHES	MM
.003	.08	.343	8.71
.005	.13	.391	9.93
.015	.39	.500	12.70
.039	.99	.625	15.88
.047	1.19	.812	20.63
.156	3.96	1.187	30.15
.250	6.35	1.594	40.49
		2.025	51.44



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The quartz resonator shall be connected between the pins indicated.

FIGURE 1. Dimensions and configuration.

Ⓔ denotes changes.

FSC 5955

Source: https://assist.dla.mil - Downloaded: 2026-10-16T16:15Z
Check the source to verify that this is the current version before use.

TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-21/U (MIL-H-10056/10)
Dimensions, marking, and configuration	See figure
Frequency range	75 to 200 kHz, inclusive
(E) Frequency tolerance	±100 PPM
Equivalent resistance	kHz: 75 to 120 — Ohms, maximum: 2,000 120+ to 150 — 2,300 150+ to 200 — 2,500
Mode of oscillation	Fundamental
Operating temperature range	-40° to +70°C, inclusive
Test set, calibration values, and rated drive level	See table II
Shock:	
(E) Frequency change permitted	±10 PPM
Equivalent-resistance change permitted	±15 percent
Vibration: 1/	
(E) Frequency change permitted	±10 PPM
Equivalent-resistance change permitted	±15 percent
(E) Thermal shock:	
Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
Bond strength:	kHz: 75 to 120 — Grams, minimum: 1,000 120+ to 160 — 800 160+ to 200 — 700

1/ Method 201 of MIL-STD-202, amplitude 0.015 inch (0.030 inch total excursion).

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	kHz	Ohms	mA	Volts	mW
TS-710/TSM	75 to 120	1400	---	1.67	2.0 ±0.4
	120+ to 150	1600	---	1.79	
	150+ to 200	1800	---	1.90	

NOTE:

1. This crystal-unit type is to be used for special applications and for replacement purposes only. Specific approval of the Government is required prior to each intended use. See MIL-STD-683.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:15Z
Check the source to verify that this is the current version before use.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Army -
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DLA - ES

(Project 5955-0512-2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:15Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-3098/2E CRYSTAL UNIT, QUARTZ, CR-16B/U

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T10:15Z
Check the source to verify that this is the current version before use.